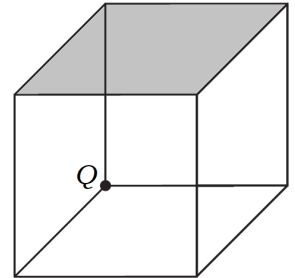


Instruction: Select as many questions as required to complete 50 points.

Q1. (10 points) A **point charge** Q sits at a corner of a cube, as shown here. What is the flux of $\mathbf{E}$ through the shaded side?

Hint: The position of charge may be considered to be at the origin. To evaluate flux in *Coulomb*, you need to solve the following equation over the specified shaded surface:

$$\psi = \epsilon_0 \int \mathbf{E} \cdot d\mathbf{S}$$

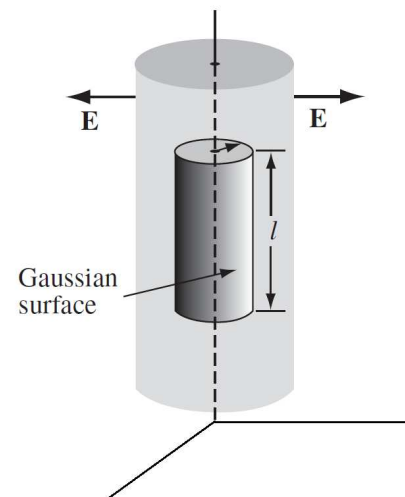


Q2. (10 points) A **long solid cylinder** carries a volumetric charge density that is proportional to the distance from the axis: $\rho_v = k\rho$, for some constant k . Using Gauss's theorem, show that the electric field $\mathbf{E}$ inside this cylinder is given as:

$$\mathbf{E} = \frac{k}{3\epsilon_0} \rho^2 \mathbf{a}_\rho$$

Hint: It is an infinite cylinder, for which $\mathbf{E} = E_\rho \mathbf{a}_\rho$ is true. You may consider a Gauss's closed surface (an inner cylinder of height l and field radius ρ , as shown here) and evaluate the following:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} Q_{\text{enclosed}}$$



Q3. (20 points) A **thick spherical shell** carries charge density

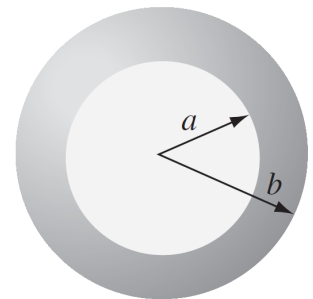
$$\rho_v = \begin{cases} \frac{k}{r^2}, & a \leq r \leq b \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the electric field $\mathbf{E}$ in the three regions:

(i) $r < a$, (ii) $a < r < b$, (iii) $r > b$.

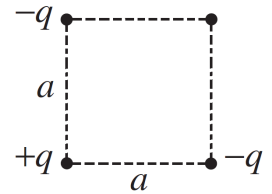
(b) Plot $|\mathbf{E}|$ as a function of r , for the case $b = 2a$.

Hint: Owing to symmetry, $\mathbf{E} = E_r \mathbf{a}_r$ is true.



Q4. (10 points)

(a) Three charges are situated at the corners of a square (each side is length a). How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?



(b) How much work does it take to assemble the whole configuration of four charges?

Q5. (10 points) Find the energy of a uniformly charged spherical (hollow) shell of total charge q and radius a .

Hint: The sphere acts like a point charge; if the total charge is q , then the field $\mathbf{E}$ outside the sphere is given as

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \mathbf{a}_r \quad (a \leq r < \infty)$$

Use $W_E = (\epsilon_0/2) \int E^2 dv$, where the volume (triple) integral is evaluated for the **entire space**; say we integrate (in the spherical coordinate system) from the surface of the sphere to infinity. We can exclude sphere's inner region because the field is zero inside.

Q6. (10 points) Determine the total charge

(a) On the cylinder $\rho = 4$, $0 < z < 10$, if $\rho_s = \rho z^3$ [nC/m²]

(b) Within the cylinder $r = 3$, if $\rho_v = \frac{10}{r^2 \sin \theta}$ [C/m³]

(c) On line $0 < x < 5$, if $\rho_l = 12x^3$ [C/m]
